

IGCSE • Cambridge (CIE) • Maths

 2 hours  27 questions

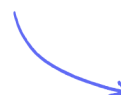
Target Test

Created 29/05/26

Total Marks

/91

Scan here for the answers
or visit [savemyexams.com](https://www.savemyexams.com)



1

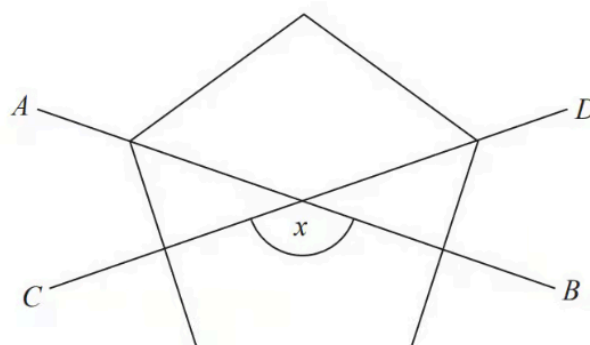


Diagram **NOT**
accurately drawn

The diagram shows a regular pentagon.

AB and CD are two of the lines of symmetry of the pentagon.

Work out the size of the angle marked x .

You must show all your working.

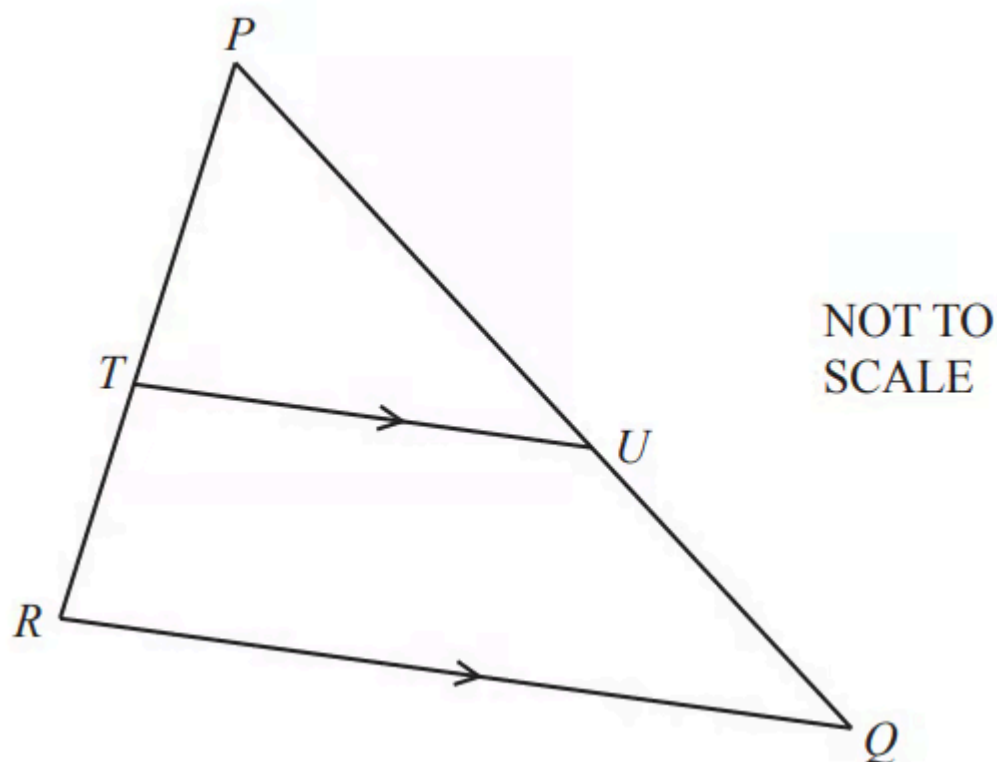
(4 marks)

2 Change a speed of 50 metres per second to a speed in kilometres per hour.

..... kilometres per hour

(3 marks)

3



PQR is a triangle.

T is a point on PR and U is a point on PQ .

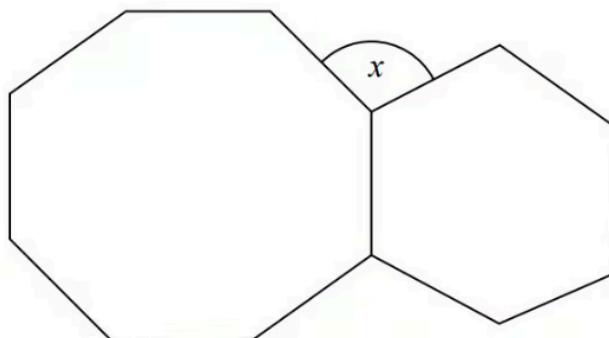
RQ is parallel to TU .

Explain why triangle PQR is similar to triangle PUT .

Give a reason for each statement you make.

(3 marks)

4



The diagram shows a regular octagon and a regular hexagon.

Find the size of the angle marked x .

You must show all your working.

(3 marks)

5 Find the number of sides of a regular polygon with interior angle 150°

(2 marks)

6 The angles of a quadrilateral are x° , $(x + 5)^\circ$, $(2x - 25)^\circ$ and $(x + 10)^\circ$.

Find the value of x .

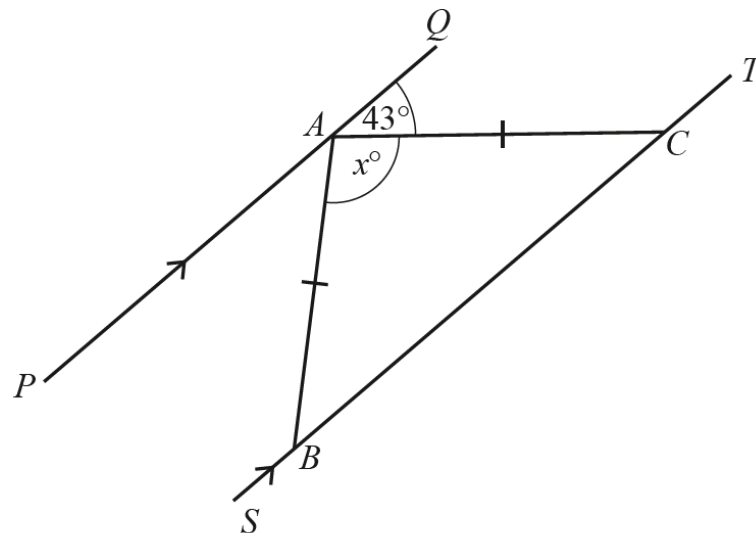
$x = \dots\dots\dots$

(3 marks)

7 Find the number of sides of a regular polygon with interior angle 162° .

(2 marks)

8



NOT TO
SCALE

The diagram shows two parallel lines PAQ and $SBCT$.

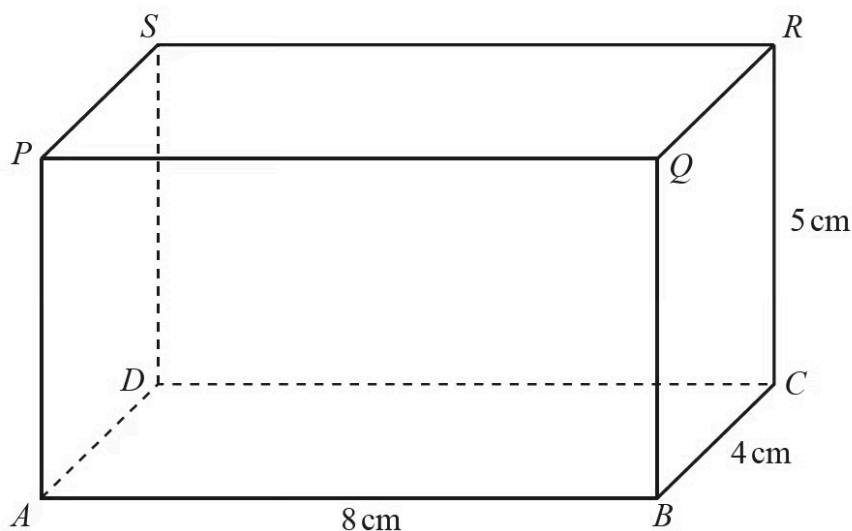
$AB = AC$ and angle $QAC = 43^\circ$.

Find the value of x .

$x = \dots\dots\dots$

(2 marks)

9



NOT TO
SCALE

The diagram shows a cuboid.

$AB = 8$ cm, $BC = 4$ cm and $CR = 5$ cm.

Write down the number of planes of symmetry of this cuboid.

(1 mark)

10 The interior angle of a regular polygon with n sides is 150° .

Calculate the value of n .

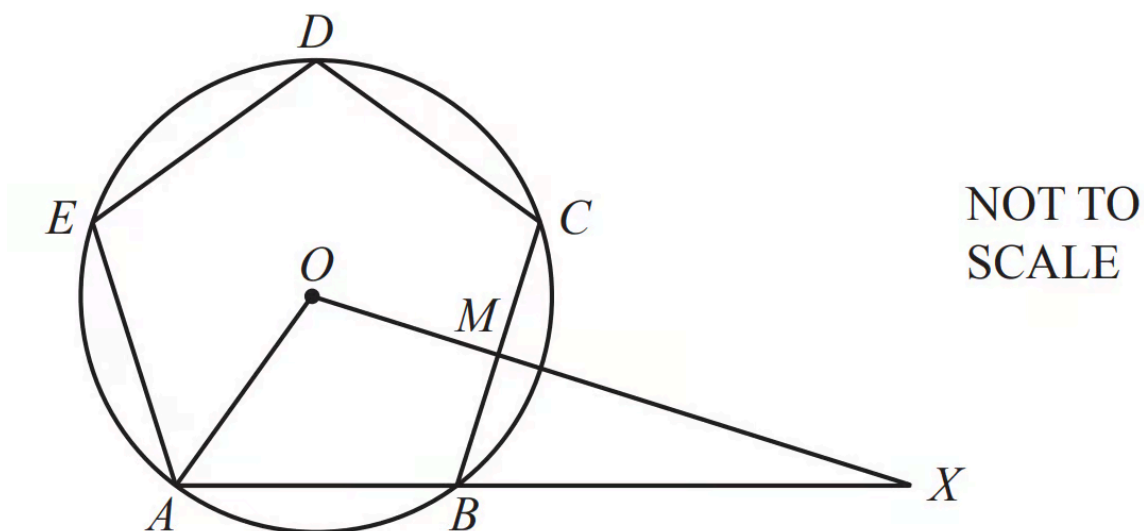
$n = \dots\dots\dots$

(2 marks)

11 (a) Show that each interior angle of a regular pentagon is 108° .

(2 marks)

(b)



The diagram shows a regular pentagon $ABCDE$.

The vertices of the pentagon lie on a circle, centre O , radius 12 cm.

M is the midpoint of BC .

i) Find BM .

$BM = \dots\dots\dots$ cm [3]

OMX and ABX are straight lines.

ii) Find BX .

$BX = \dots\dots\dots$ cm [3]

ii) Calculate the area of triangle AOX .

$\dots\dots\dots$ cm² [3]

(9 marks)

- 12 The diagram shows a regular 10-sided polygon, $ABCDEFGHIJ$

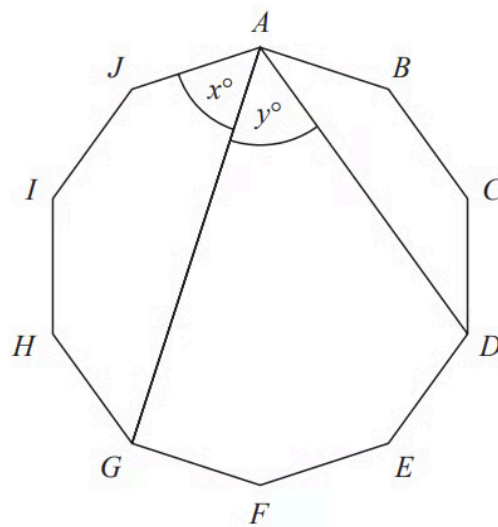
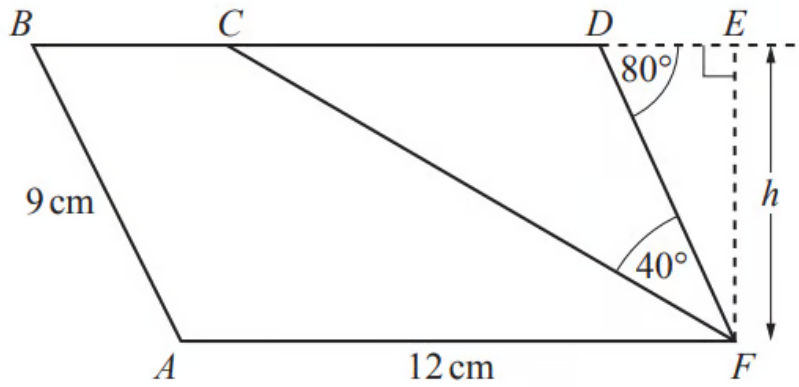


Diagram **NOT**
accurately drawn

Show that $x = y$.

(4 marks)

13



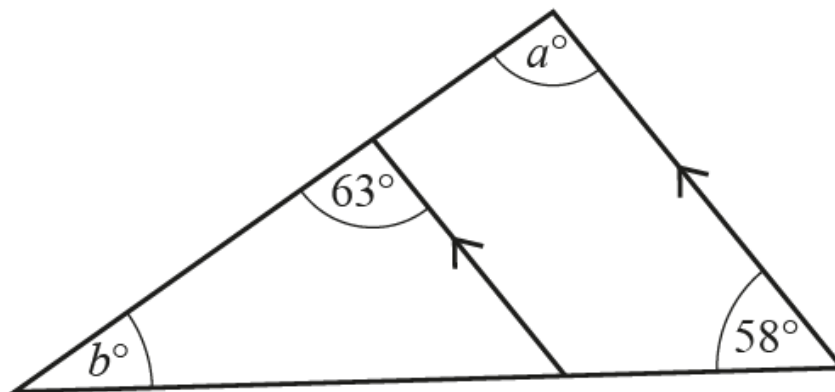
NOT TO
SCALE

$ABDF$ is a parallelogram and $BCDE$ is a straight line.
 $AF = 12\text{ cm}$, $AB = 9\text{ cm}$, angle $CFD = 40^\circ$ and angle $FDE = 80^\circ$.

Explain why triangle CDF is isosceles.

(2 marks)

14



NOT TO
SCALE

Complete the statements.

$a = \dots\dots\dots$ because $\dots\dots\dots$

$b = \dots\dots\dots$ because $\dots\dots\dots$

(4 marks)

15

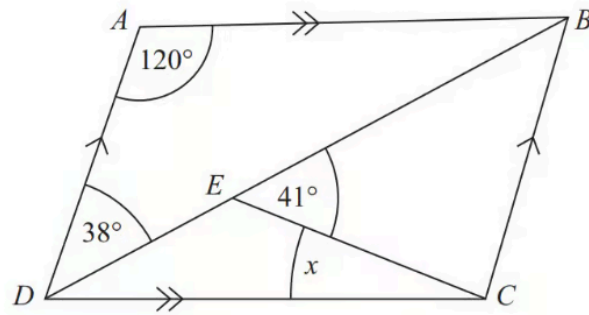


Diagram **NOT**
accurately drawn

$ABCD$ is a parallelogram.

Angle $ADB = 38^\circ$.

Angle $BEC = 41^\circ$.

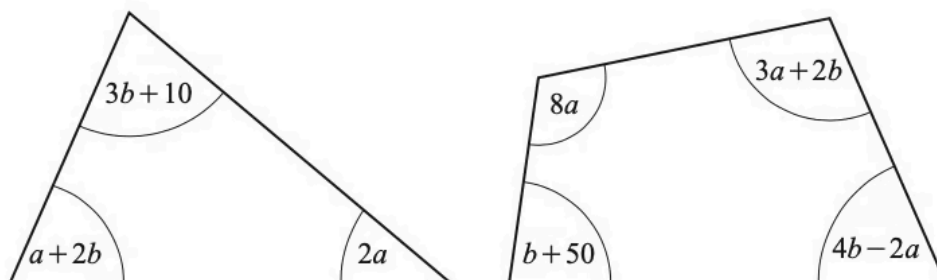
Angle $DAB = 120^\circ$.

Calculate the size of angle x .

You must give reasons for your answer.

(4 marks)

16 The diagram shows a triangle and a quadrilateral. All angles are in degrees.



NOT TO
SCALE

i) For the triangle, show that $3a + 5b = 170$.

[1]

ii) For the quadrilateral, show that $9a + 7b = 310$.

[1]

iii) Solve these simultaneous equations.
Show all your working.

$a = \dots\dots\dots$

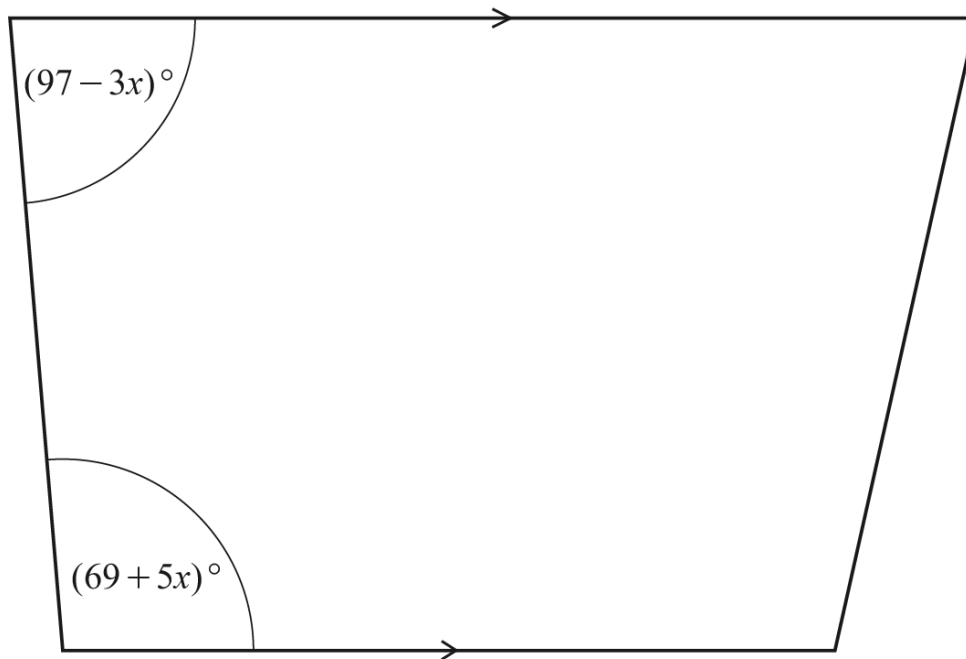
$b = \dots\dots\dots$ [3]

iv) Find the size of the smallest angle in the triangle.

[1]

(6 marks)

- 17 The diagram shows a trapezium.



NOT TO
SCALE

Work out the value of x .

$x =$

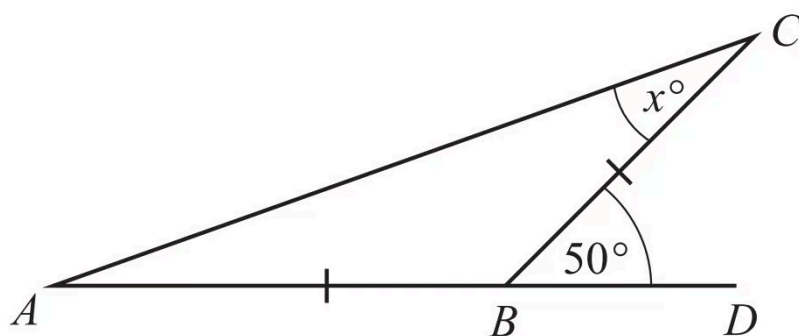
(3 marks)

- 18 Change 457000 cm^2 into m^2 .

..... m^2

(1 mark)

19



NOT TO
SCALE

$AB = BC$ and ABD is a straight line.

Find the value of x .

$x = \dots\dots\dots$

(2 marks)

20 The lake behind a dam has an area of 55 hectares.

When the gates in the dam are open, water flows out at a rate of 75 000 litres per second.

i) Show that 90 million litres of water flows out in 20 minutes.

[1]

ii) Beneath the surface, the lake has vertical sides.

Calculate the drop in the water level of the lake when the gates are open for 20 minutes.
Give your answer in centimetres.

[1 hectare = 10^4 m^2 , 1000 litres = 1 m^3]

$\dots\dots\dots$ cm [3]

(4 marks)

21 The interior angle of a regular polygon with n sides is 156° .

Work out the value of n .

$n = \dots\dots\dots$

(2 marks)

22 In a regular polygon, the interior angle is 11 times the exterior angle.

i) Work out the number of sides of this polygon.

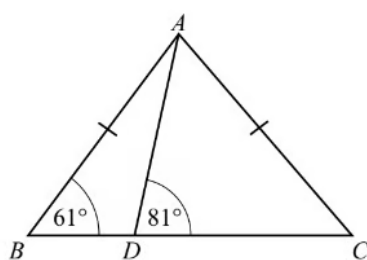
[3]

ii) Find the sum of the interior angles of this polygon.

[2]

(5 marks)

- 23 The diagram shows two triangles, ABD and ADC .



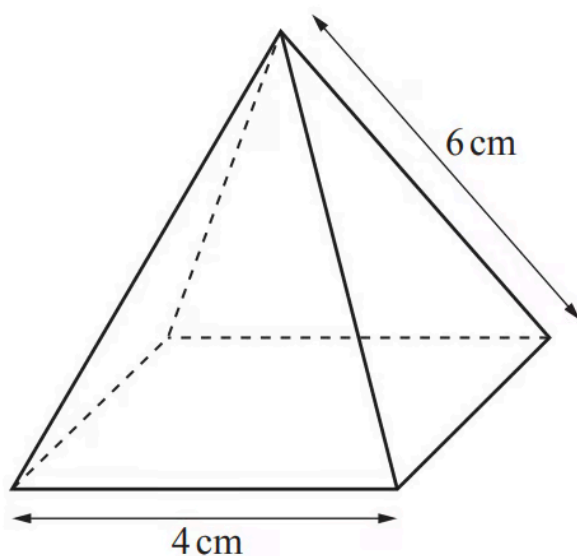
NOT TO
SCALE

BDC is a straight line, $AB = AC$, angle $ABD = 61^\circ$ and angle $ADC = 81^\circ$. Work out angle DAC .

Angle $DAC = \dots\dots\dots$

(2 marks)

24



NOT TO
SCALE

The diagram shows a pyramid with a square base.
The triangular faces are congruent isosceles triangles.

Write down the number of planes of symmetry of this pyramid.

(1 mark)

25 $ABCDEFGHI$ is a regular 9-sided polygon.

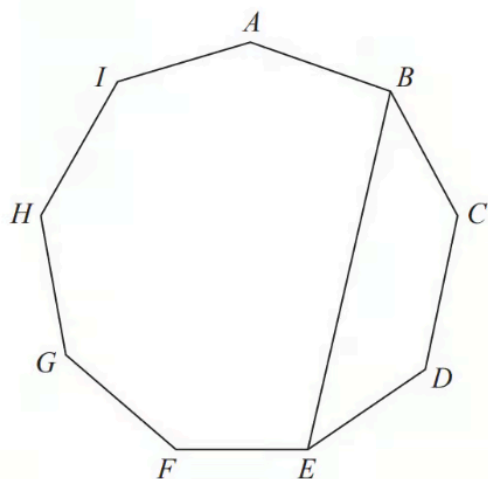


Diagram **NOT**
accurately drawn

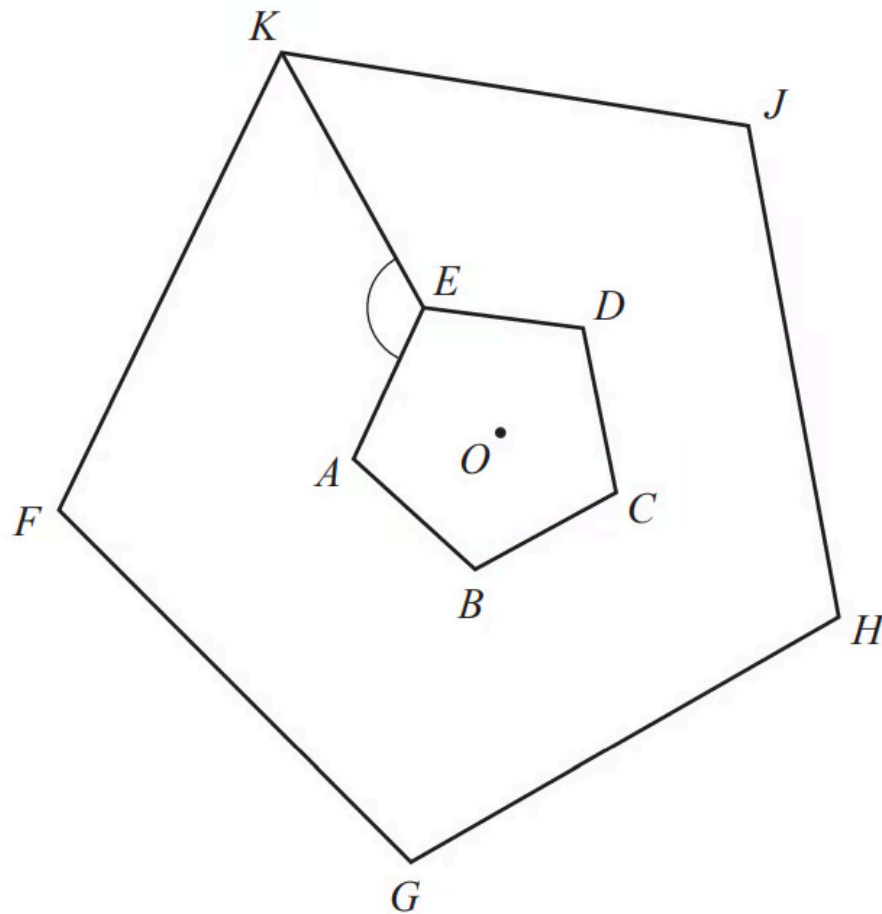
The vertices B and E are joined with a straight line.

Work out the size of angle BEF .

You must show how you get your answer.

(4 marks)

26 (a)



NOT TO
SCALE

The diagram shows two regular pentagons.

Pentagon $FGHIK$ is an enlargement of pentagon $ABCDE$, centre O .

Find angle AEK .

Angle AEK =

(4 marks)

(b) The area of pentagon $FGHIK$ is 73.5 cm^2 .

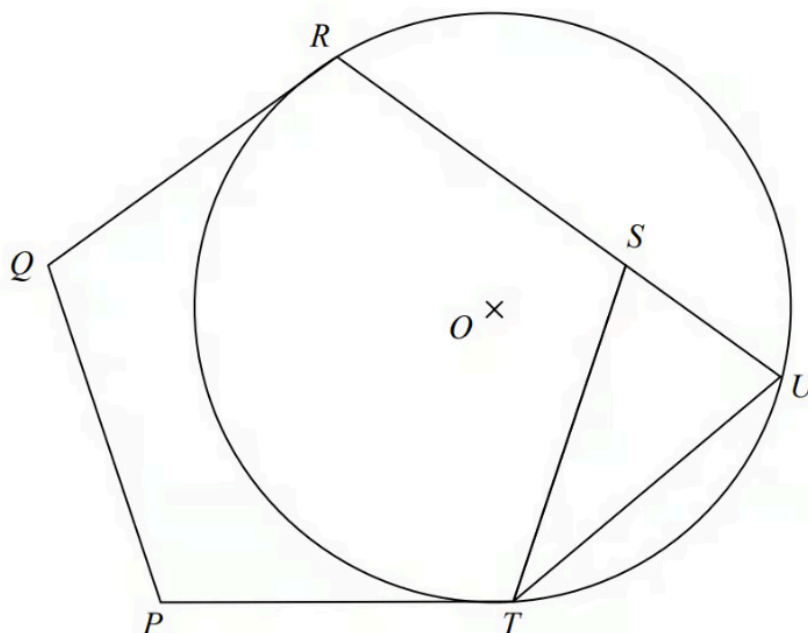
The area of pentagon $ABCDE$ is 6 cm^2 .

Find the ratio perimeter of pentagon $FGHIK$: perimeter of pentagon $ABCDE$ in its simplest form.

..... :

(2 marks)

27



$PQRST$ is a regular pentagon.
 R , U and T are points on a circle, centre O .
 QR and PT are tangents to the circle.
 RSU is a straight line.

Prove that $ST = UT$.

(5 marks)